

Moons & Rings of Jupiter

An OpenSCAD 3D Model

Technical Document

Source files:

JovianSystem.scad
ModelOfJupiter.scad
DetailedModelOfJupiter.scad
MajorMoonsOfJupiter.scad
SmallerMoonsOfJupiter.scad

This document describes the mathematical framework, physical concepts, geometric construction techniques, and overall architecture of the OpenSCAD codebase that models the Jovian system — Jupiter itself, its four ring families, its Galilean satellites, its inner shepherd moons, and its large population of irregular outer moons — together with associated physical phenomena such as the magnetosphere, plasma torus, radiation belts, aurorae, and tidal interaction lines.

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1 Introduction

Moons & Rings of Jupiter is a static, parameterised 3-D model of the Jovian system constructed entirely in OpenSCAD — a script-based, CSG (Constructive Solid Geometry) solid-modelling application. The project spans five source files of increasing complexity and visual richness:

- **JovianSystem.scad** — a compact overview scene that places all principal bodies in one coordinate space with simplified geometry.
- **ModelOfJupiter.scad** — a high-fidelity, procedurally coloured model of Jupiter’s surface based on multi-scale trigonometric field synthesis.
- **DetailedModelOfJupiter.scad** — the most feature-complete file, adding axial tilt, rotation axis, magnetic-field geometry, magnetosphere, plasma torus, radiation belts, aurorae, Roche limit, ring system, animated Galilean moon orbits under the Laplace resonance, and tidal interaction lines.
- **MajorMoonsOfJupiter.scad** — Jupiter plus the four classes of named moons (inner, Galilean, and two irregular satellites) with parameterised orbit paths, labels, and orbital periods.
- **SmallerMoonsOfJupiter.scad** — Jupiter plus a large catalogue of smaller and provisional moons, each with an individually constructed and inclined orbit path.

All models are *static and deterministic*: no simulation engine is involved; every visual feature is generated by pure mathematical functions evaluated per surface patch or geometric primitive.

2 Components of the Jovian System

2.1 Jupiter

Jupiter is the largest planet in the Solar System, a gas giant with an equatorial radius of approximately 71,492 km. In every file it is rendered as a sphere of model radius $R = 40$ (arbitrary model units). The base model uses a colour-per-face tessellation strategy: the spherical surface is divided into $N_\phi \times N_\lambda$ latitude-longitude patches, each assigned a procedurally computed RGB colour that encodes the complex banded structure of the Jovian atmosphere (Section ??).

Key physical features modelled on the surface include:

- **Equatorial belt–zone system** (dark belts alternating with bright zones, spanning ≈ -35 to $+35$ latitude).
- **Dark and bright filaments** (narrow sub-belt cloud streams).
- **Multi-scale atmospheric noise** (large, medium, fine scale).
- **Polar regions** (concentric broken ring structures, filaments, crossflow, vortices, and a central polar cyclone).
- **Great Red Spot** (GRS), an anticyclonic superstorm centred near 22 S, 20 W.

- **Small atmospheric storms** scattered across mid-latitudes.
- **Auroral ovals** (in `DetailedModelOfJupiter.scad`).
- **North–south brightness asymmetry** (southern hemisphere rendered slightly darker).

2.2 Ring System

Jupiter possesses four concentric ring families, all modelled as thin, semi-transparent annular discs constructed via cylinder-difference operations or inclined orbit-ring cylinders.

Table 1: Jupiter’s ring system as modelled.

Ring	Source body	Model inner r	Model outer r
Halo ring	—	$\approx 1.05R$	$\approx 1.18R$
Main ring	Metis, Adrastea	$\approx 1.15R$	$\approx 1.26R$
Amalthea gossamer	Amalthea	$\approx 1.30R$	$\approx 1.52R$
Thebe gossamer	Thebe	$\approx 1.52R$	$\approx 1.82R$

In `JovianSystem.scad` the rings are modelled as flat cylinders with slight rotational tilts (to simulate finite ring-plane inclination), whereas in `DetailedModelOfJupiter.scad` and `MajorMoonsOfJupiter.scad` a `difference()` of two coaxial cylinders produces a proper annular disc.

2.3 Galilean Moons

The four Galilean satellites — Io, Europa, Ganymede, and Callisto — discovered by Galileo Galilei in 1610, are the largest of Jupiter’s moons. They are individually coloured to reflect their real surface chemistry:

Table 2: Galilean moon properties as represented in the models.

Moon	Colour (hex approx.)	Surface character	Period (real)	Resonan
Io	#FFD96D (sulphurous yellow)	Volcanic sulphur plains	1.769 d	4
Europa	#E2E3E3 (icy white)	Water-ice crust with lineae	3.551 d	2
Ganymede	#8F8071 (grey-brown)	Mixed icy/rocky terrain	7.155 d	1
Callisto	#B4A285 (dark brown)	Heavily cratered ice	16.69 d	0.427

2.4 Inner (Shepherd) Moons

Four small inner moons orbit within or at the edge of the ring system: Metis, Adrastea (main-ring shepherd moons), Amalthea, and Thebe. They are represented as small spheres with their orbital inclinations encoded via the `rotate()` transform applied to each orbit path.

2.5 Irregular Outer Moons

`MajorMoonsOfJupiter.scad` models four irregular moons: Himalia, Elara (prograde Himalia group), and Pasiphae and Ananke (retrograde groups). Their retrograde orbits are indicated by positioning the moons on the negative- x side of Jupiter.

`SmallerMoonsOfJupiter.scad` extends this to include dozens of smaller and provisional moons, including Euporie, Eupheme, Mneme, Euanthe, Harpalyke, Orthosie, Helike, and several provisional designations (e.g. S/2003 J 18, S/2010 J 2, S/2016 J 1, S/2021 J 3). Each receives an individually inclined orbit ring constructed from the same segmented-polygon technique.

2.6 Physical Phenomena (`DetailedModelOfJupiter.scad`)

The detailed model renders several additional physical structures:

Rotation axis A red rod and arrowhead showing Jupiter’s 3.13 axial tilt, labelled with the rotation period $P = 9\text{ h }56\text{ min}$.

Equatorial and ecliptic planes Translucent horizontal discs defining the tilt geometry.

Rotation arrow A partial `rotate_extrude` arc indicating the prograde (westward) spin direction.

Magnetic field lines Dipole field-line curves parameterised by L-shell value and azimuthal angle (Section ??).

Magnetosphere A sunward-compressed, tail-elongated oblate sphere.

Io plasma torus A toroidal purple-red glow centred on Io’s orbit, produced by volcanic sulphur and oxygen ions.

Radiation belts Three concentric toroidal intensity zones (inner, middle, outer) analogous to Jupiter’s Van Allen belts.

3-D aurorae Conical glowing rings at both poles with a red outer edge, pulsed by the OpenSCAD animation parameter `$t`.

Roche limit A dotted spherical surface at $r = 1.75R$ (the approximate tidal disruption radius).

Tidal interaction lines Dashed radial lines co-rotating with each Galilean moon.

3 Mathematical Framework

3.1 Utility Functions

3.1.1 Clamping

All colour-weight computations are bounded to the interval $[0, 1]$ using the clamp function:

$$\text{clamp}(x, 0, 1) = \max(0, \min(1, x)) \quad (1)$$

3.1.2 Cubic Smoothstep Interpolation

The smoothstep function produces a smooth transition between 0 and 1 with zero first derivatives at both endpoints, eliminating the sharp boundaries introduced by linear clamping:

$$S(x) = q^2(3 - 2q), \quad q = \text{clamp}(x, 0, 1) \quad (2)$$

Expanding:

$$S(x) = 3q^2 - 2q^3 \quad (3)$$

Key properties:

$$S(0) = 0, \quad S(1) = 1, \quad (4)$$

$$S'(q) = 6q(1 - q), \quad S'(0) = S'(1) = 0. \quad (5)$$

3.1.3 Linear (Vector) Interpolation

Colour blending between two RGB vectors $\mathbf{a}$ and $\mathbf{b}$ by weight t :

$$\text{mix}(\mathbf{a}, \mathbf{b}, t) = \mathbf{a} + q(\mathbf{b} - \mathbf{a}) = (1 - q)\mathbf{a} + q\mathbf{b}, \quad q = \text{clamp}(t, 0, 1) \quad (6)$$

Component-wise: $C_i(t) = a_i + q(b_i - a_i)$.

3.1.4 Periodic Longitude Wrapping

Longitude λ is periodic with period 360. The wrap function maps any value back to $[-180, 180]$:

$$\text{wrap}(\lambda) = \begin{cases} \lambda - 360 & \lambda > 180, \\ \lambda + 360 & \lambda < -180, \\ \lambda & \text{otherwise.} \end{cases} \quad (7)$$

3.2 Spherical Coordinate System

Every surface point on Jupiter is parameterised by geodetic latitude $\varphi \in [-90, 90]$ and longitude $\lambda \in [-180, 180)$. The Cartesian position of a point on a sphere of radius R is:

$$\mathbf{P}(\varphi, \lambda) = \begin{pmatrix} R \cos \varphi \cos \lambda \\ R \cos \varphi \sin \lambda \\ R \sin \varphi \end{pmatrix} \quad (8)$$

This satisfies the sphere equation $x^2 + y^2 + z^2 = R^2$ as can be verified:

$$x^2 + y^2 = R^2 \cos^2 \varphi (\cos^2 \lambda + \sin^2 \lambda) = R^2 \cos^2 \varphi, \quad (9)$$

$$z^2 = R^2 \sin^2 \varphi, \quad (10)$$

$$x^2 + y^2 + z^2 = R^2 (\cos^2 \varphi + \sin^2 \varphi) = R^2. \quad (11)$$

3.3 Fourier-like Trigonometric Noise

Real atmospheric textures are approximated by finite trigonometric series (analogous to truncated Fourier expansions) evaluated at each surface point (φ, λ) . Three scale levels are defined.

3.3.1 Large-Scale Noise

$$N_L(\varphi, \lambda) = 0.42 \sin(1.25\lambda + 0.80\varphi) + 0.27 \cos(2.70\lambda - 1.10\varphi) + 0.16 \sin(5.20\lambda + 2.20\varphi) + 0.10 \cos(8.70\lambda - 3.40\varphi) \quad (12)$$

3.3.2 Medium-Scale Noise

$$N_M(\varphi, \lambda) = 0.42 \sin(4.10\lambda + 2.00\varphi) + 0.27 \cos(8.70\lambda - 3.70\varphi) + 0.17 \sin(15.0\lambda + 5.20\varphi) + 0.09 \cos(27.0\lambda - 8.00\varphi) \quad (13)$$

3.3.3 Fine-Scale Noise

$$N_F(\varphi, \lambda) = 0.38 \sin(12.0\lambda + 5.0\varphi) + 0.26 \cos(23.0\lambda - 8.0\varphi) + 0.18 \sin(39.0\lambda + 13.0\varphi) + 0.10 \cos(67.0\lambda - 21.0\varphi) \quad (14)$$

All three fields have the general form of a finite Fourier-like series:

$$N(\varphi, \lambda) = \sum_k A_k f_k(\alpha_k \lambda + \beta_k \varphi + \phi_k), \quad f_k \in \{\sin, \cos\} \quad (15)$$

Higher spatial frequencies α_k correspond to shorter wavelengths $\lambda_k \propto 2\pi/\alpha_k$, generating progressively finer atmospheric texture.

3.4 Wind-Wave Displacement

Jupiter's jet streams produce latitudinal oscillations of atmospheric bands. These are modelled by the wind-wave function, a four-term Fourier sum in longitude with a phase parameter ϕ :

$$W(\lambda, \phi) = 3.2 \sin(1.35\lambda + \phi) + 2.0 \sin(2.70\lambda - 0.70\phi) + 1.15 \sin(5.20\lambda + 1.40\phi) + 0.55 \sin(9.70\lambda - 1.90\phi) \quad (16)$$

Different frequencies interfere through superposition, producing an aperiodic-looking but fully deterministic longitudinal modulation.

3.5 Atmospheric Ribbon (Gaussian Band)

Each atmospheric belt or zone is a *ribbon*: a Gaussian-shaped intensity profile centred on a latitude that oscillates with longitude.

Step 1 — Shifted centre:

$$C'(\lambda) = C_0 + A \sin(f\lambda + \phi) + 0.35 W(\lambda, \phi) \quad (17)$$

Step 2 — Latitudinal displacement:

$$\delta(\varphi, \lambda) = \varphi - C'(\lambda) \quad (18)$$

Step 3 — Gaussian intensity:

$$R(\varphi, \lambda) = \exp \left[- \left(\frac{\delta}{w} \right)^2 \right] \quad (19)$$

At $\delta = 0$ the ribbon has its maximum intensity $R = 1$; as $|\delta| \rightarrow \infty$, $R \rightarrow 0$. The parameter w controls the effective width of the atmospheric band.

The complete dark-belt field is a weighted superposition of 14 individual ribbons:

$$D(\varphi, \lambda) = \sum_{i=1}^{14} w_i R_i(\varphi, \lambda) \quad (20)$$

Similarly for bright belts (9 ribbons), dark filaments (12 ribbons, narrow $w \approx 0.3$ – 0.42), and bright filaments (8 ribbons).

3.6 Polar Geometry

3.6.1 Polar Distance

The angular distance from the nearest pole is:

$$p(\varphi) = 90 - |\varphi| \quad (21)$$

Therefore $p = 0$ at the poles ($\varphi = \pm 90$) and $p = 90$ at the equator ($\varphi = 0$).

3.6.2 Polar Blend

The transition between the equatorial colour model and the polar colour model is controlled by:

$$B_p(\varphi) = S\left(\text{clamp}\left(\frac{52-p}{16}, 0, 1\right)\right) \quad (22)$$

where S is the smoothstep of Eq. (??). The blend begins near $p = 52$ ($|\varphi| \approx 38$) and is complete at $p = 36$ ($|\varphi| \approx 54$).

3.6.3 Polar Warp

A multi-frequency perturbation field creates irregular polar ring geometry:

$$W_p(\lambda, p, \phi) = 3.2 \sin(\lambda + \phi) + 1.9 \sin(2\lambda - 1.7\phi) + 1.15 \sin(3\lambda + 0.7p + \phi) + 0.75 \sin(5\lambda - 1.2p) + 0.40 \sin(9\lambda + 2p) \quad (23)$$

3.6.4 Polar Microstructure

High-frequency oscillations add fine-grained texture inside the polar cap:

$$M_p(\lambda, p) = 0.75 \sin(11\lambda + 3p) + 0.48 \sin(19\lambda - 4.5p) + 0.32 \sin(31\lambda + 7p) + 0.19 \sin(47\lambda - 10p) + 0.10 \sin(71\lambda - 13p) \quad (24)$$

3.6.5 Polar Ring

A Gaussian-shaped concentric ring in the polar cap, with a varying width:

$$C_r(\lambda, p, \phi, s) = r_0 + s W_p(\lambda, p, \phi) + 0.35 M_p(\lambda, p) \quad (25)$$

$$w_v(\lambda, \phi) = w[0.72 + 0.28(0.5 + 0.5 \sin(3\lambda + \phi))] \quad (26)$$

$$R_p(\varphi, \lambda) = \exp\left[-\left(\frac{p - C_r}{w_v}\right)^2\right] \quad (27)$$

3.6.6 Polar Ring Breaking

An oscillatory modulation field fragments the polar rings:

$$n(\lambda, \phi) = 0.45 \sin(\lambda + \phi) + 0.30 \sin(2\lambda - 1.3\phi) + 0.20 \sin(4\lambda + 2.2\phi) + 0.12 \sin(7\lambda - \phi) \quad (28)$$

$$Q(\lambda, \phi) = 0.20 + 0.80 S\left(\text{clamp}\left(\frac{n + 0.08}{0.35}, 0, 1\right)\right) \quad (29)$$

The broken ring field is then the pointwise product:

$$B_r(\varphi, \lambda) = R_p(\varphi, \lambda) \cdot Q(\lambda, \phi) \quad (30)$$

Since $Q \geq 0.20$, the ring never vanishes completely. Nine concentric broken rings ($r_0 = 8, 13, 17.5, 22.5, 27.5, 33, 38, 43, 47.5$) build up the polar band field.

3.6.7 Polar Vortices

Individual polar cyclones are localised using an approximately Euclidean distance in polar coordinates. For a vortex centred at polar radius r_0 and longitude λ_0 :

$$\Delta\lambda = \text{wrap}(\lambda - \lambda_0) \quad (31)$$

$$x_v = \Delta\lambda \cos r_0 \quad (32)$$

$$y_v = p - r_0 \quad (33)$$

$$d_v = \sqrt{x_v^2 + y_v^2} \quad (34)$$

$$V(\varphi, \lambda) = s \exp\left[-\left(\frac{d_v}{\sigma}\right)^2\right] \quad (35)$$

The cosine factor in Eq. (??) approximately accounts for the convergence of meridians near the pole. A total of 22 vortices are superimposed for the polar vortex field.

3.6.8 Polar Crossflow

Nested non-linear oscillations simulate the transverse atmospheric flow across the polar cap:

$$a = \sin(3\lambda + 7p + 2\sin(2\lambda)) \quad (36)$$

$$b = \sin(6\lambda - 4p + \sin(5\lambda)) \quad (37)$$

$$c = \sin(13\lambda + 9p), \quad d = \sin(23\lambda - 15p) \quad (38)$$

$$C_f(\varphi, \lambda) = 0.45a + 0.28b + 0.17c + 0.10d \quad (39)$$

The inner sine terms in a and b introduce *phase modulation*, producing more complex structure than a plain Fourier sum.

3.7 Storm Modelling

Small atmospheric storms use a normalised elliptical distance metric:

$$d_s = \sqrt{\left(\frac{\varphi - \varphi_c}{a}\right)^2 + \left(\frac{\text{wrap}(\lambda - \lambda_c)}{b}\right)^2} \quad (40)$$

$$S_s(\varphi, \lambda) = \text{smooth}(1 - d_s) \quad (41)$$

The storm field is maximum at its centre ($d_s = 0$) and decays to zero for $d_s \geq 1$. Eight storms are superimposed. Storms are wider in longitude than latitude ($b > a$), reflecting the real elongation of Jovian convective storms.

3.8 Great Red Spot

The Great Red Spot (GRS) is a long-lived anticyclonic storm centred at approximately 22 S, 20 W. It is modelled as an ellipse wider in longitude than latitude:

$$d_{\text{GRS}} = \sqrt{\left(\frac{\text{wrap}(\lambda + 20)}{18}\right)^2 + \left(\frac{\varphi + 22}{7.8}\right)^2} \quad (42)$$

$$M_{\text{GRS}}(\varphi, \lambda) = \text{smooth}(1 - d_{\text{GRS}}) \quad (43)$$

Internal texture is provided by a multi-frequency field:

$$T_{\text{GRS}} = 0.42 + 0.23 \sin(8\lambda + 12\varphi) + 0.17 \cos(17\lambda - 9\varphi) + 0.10 \sin(29\lambda + 21\varphi) + 0.08 \cos(47\lambda - 32\varphi) \quad (44)$$

The GRS colour interpolates from a dark reddish-brown to a lighter red using T_{GRS} as the interpolation weight, then blends over the base atmosphere with weight $0.88 M_{\text{GRS}}$.

3.9 Jupiter Surface Colour Synthesis Pipeline

The complete colour at any point (φ, λ) is computed by a sequential linear-interpolation chain, building up from a cream base through eleven stages:

1. $D = \text{clamp}(0.58 \mathbf{D})$, $B = \text{clamp}(0.52 \mathbf{B})$, $D_f = \text{clamp}(0.52 \mathbf{D}_f)$, $B_f = \text{clamp}(0.38 \mathbf{B}_f)$
where $\mathbf{D}, \mathbf{B}, \mathbf{D}_f, \mathbf{B}_f$ are the dark-belt, bright-belt, dark-filament, and bright-filament fields.
2. $\mathbf{C}_0 = \text{mix}(\text{CREAM}, \text{BROWN}, D)$ (Base: cream–brown blend driven by dark-belt coverage.)
3. $w_{\text{warm}} = \text{clamp}(0.65D - 0.12N_L)$, $\mathbf{C}_1 = \text{mix}(\mathbf{C}_0, \text{OCHRE}, 0.35w_{\text{warm}})$ (Warm tones in darker-belt regions.)
4. $\mathbf{C}_2 = \text{mix}(\mathbf{C}_1, \text{DARK}, 0.42D_f)$ (Dark filament overlays.)
5. $\mathbf{C}_3 = \text{mix}(\mathbf{C}_2, \text{LIGHT}, 0.42B)$ (Bright belt highlights.)
6. $\mathbf{C}_4 = \text{mix}(\mathbf{C}_3, \text{WHITE}, 0.52B_f)$ (Bright filament highlights.)
7. $\mathbf{C}_5 = \text{mix}(\mathbf{C}_4, \text{TAN}, \text{clamp}((-N_M + 0.5) \cdot 0.11))$ (Medium-scale mottling.)
8. $\mathbf{C}_6 = \text{mix}(\mathbf{C}_5, \text{WHITE}, 0.13 S((N_F - 0.40)/0.60))$ (Fine-scale brightening.)
9. $\mathbf{C}_7 = \text{mix}(\mathbf{C}_6, [0.53, 0.34, 0.22], 0.18 \text{clamp}(S_{\text{storms}}))$ (Storm darkening.)
10. $\mathbf{C}_8 = \text{mix}(\mathbf{C}_7, \mathbf{C}_{\text{polar}}, B_p)$ (Polar region blend.)
11. $\mathbf{C}_{\text{final}} = \text{applyGRS}(\mathbf{C}_8)$ (Great Red Spot overlay.)

`DetailedModelOfJupiter.scad` adds a twelfth step for a latitudinal temperature-zone gradient and additional polar features (aurorae, cyclones, turbulence, spiral arms).

3.10 Orbit Construction Geometry

Orbit paths for moons are approximated as regular polygons. For a circular orbit of model radius r approximated by N line segments:

$$\text{Circumference : } C = 2\pi r \quad (45)$$

$$\text{Segment length : } \ell = \frac{C}{N} = \frac{2\pi r}{N} \quad (46)$$

$$\text{Angular step : } \Delta\alpha = \frac{360}{N} \quad (47)$$

Each segment is a small rectangle of width ℓ and thickness t , rotated by $i\Delta\alpha$ and translated to radius r . At $N = 180\text{--}500$ segments the visual result is indistinguishable from a smooth circle.

4 Physics Concepts

4.1 Jovian Atmospheric Banding

Jupiter's visible surface consists of alternating dark *belts* and bright *zones* caused by large-scale convective circulation cells coupled with the planet's rapid rotation ($P \approx 9.9\text{ h}$). The Coriolis force produces east–west zonal jets, confining convective upwelling into the bright zones and downwelling into the dark belts. The model encodes this as 14 dark and 9 bright ribbon fields (Section ??).

4.2 Axial Tilt

Jupiter's rotational axis is tilted by:

$$\varepsilon = 3.13 \quad (48)$$

relative to its orbital (ecliptic) plane. This is among the smallest axial tilts in the Solar System. In `DetailedModelOfJupiter.scad` all co-rotating components (surface, rings, moons, magnetic field) are wrapped in a `rotate([AXIAL_TILT, 0, 0])` transform, while the magnetosphere boundary and Roche limit are rendered in the ecliptic frame (untilted).

4.3 Orbital Mechanics and Kepler's Laws

The orbital radius, angular velocity, and period of each moon follow Kepler's third law:

$$T^2 = \frac{4\pi^2}{GM_J} a^3 \quad (49)$$

where G is the gravitational constant, M_J the mass of Jupiter, a the semi-major axis, and T the orbital period. The model stores real orbital periods as text labels (e.g. $T_{\text{Io}} = 1.77\text{ d}$, $T_{\text{Eu}} = 3.55\text{ d}$, etc.) and animated orbital angles are derived from the OpenSCAD parameter `$t`.

4.4 Laplace Mean-Motion Resonance

The three innermost Galilean moons satisfy the Laplace resonance, a near-exact commensurability of mean orbital angular velocities:

$$n_{\text{Io}} : n_{\text{Eu}} : n_{\text{Ga}} = 4 : 2 : 1 \quad (50)$$

or equivalently:

$$\frac{1}{T_{\text{Io}}} - \frac{3}{T_{\text{Eu}}} + \frac{2}{T_{\text{Ga}}} = 0 \quad (51)$$

In `DetailedModelOfJupiter.scad` the animated orbital angles encode this exactly:

$$\alpha_{\text{Io}} = 4 \cdot 360 \cdot \$t, \quad \alpha_{\text{Eu}} = 2 \cdot 360 \cdot \$t, \quad \alpha_{\text{Ga}} = 1 \cdot 360 \cdot \$t \quad (52)$$

The resonance maintains eccentricity in Io's and Europa's orbits via repeated gravitational pumping, which drives the intense tidal heating responsible for Io's volcanism.

4.5 Magnetic Dipole Field

Jupiter has the strongest planetary magnetic field in the Solar System. To first approximation, the external field is that of a magnetic dipole whose field lines satisfy:

$$r = L R_J \sin^2 \theta \quad (53)$$

where r is the radial distance from Jupiter's centre, θ is the colatitude (polar angle), L is the *L-shell* (McIlwain) parameter (dimensionless, equal to the equatorial crossing distance in units of R_J), and R_J is Jupiter's radius.

Converting to Cartesian coordinates for a field line of L-shell L at azimuthal angle ϕ :

$$x = L R_J \sin^2 \theta \sin \theta \cos \phi \quad (54)$$

$$y = L R_J \sin^2 \theta \sin \theta \sin \phi \quad (55)$$

$$z = L R_J \sin^2 \theta \cos \theta \quad (56)$$

In the model, field lines are drawn at $L \in \{2.0, 3.2, 4.8\}$ and $\phi \in \{0, 60, \dots, 300\}$, with θ varying from 8 to 172.

A magnetotail is separately rendered as a series of curved polylines swept in the anti-sunward ($-x$) direction.

4.6 Roche Limit

Inside the Roche limit, tidal forces from Jupiter overcome the self-gravity of any satellite, preventing accretion or tearing apart existing bodies. The Roche limit for a fluid satellite is:

$$d_R = 2.44 R_J \left(\frac{\rho_J}{\rho_m} \right)^{1/3} \quad (57)$$

where ρ_J and ρ_m are the densities of Jupiter and the satellite, respectively. For an approximate rigid-body Roche radius the factor 2.44 is replaced by ~ 1.26 . The model places the Roche limit at:

$$r_R = 1.75 R \quad (\text{model units}) \quad (58)$$

which corresponds to $\approx 70\,000$ km at real scale, consistent with Jupiter's observed ring system lying mostly inside this boundary.

4.7 Io Plasma Torus

Io's intense tidal heating drives volcanic plumes that inject approximately 10^3 kg/s of sulphur dioxide and sulphur into space. These molecules are ionised by solar UV and Jupiter's magnetosphere, forming a dense donut-shaped plasma torus co-orbiting with Io at $r \approx 5.9 R_J$.

In the model, the torus is rendered as a `rotate_extrude` of an elliptically scaled circle:

$$\text{Torus surface: } \mathbf{P}(\theta, \phi) = ((r_0 + a \cos \phi) \cos \theta, (r_0 + a \cos \phi) \sin \theta, c a \sin \phi) \quad (59)$$

where r_0 is the torus major radius (set to `ORBIT_IO` = 80 model units), a is the tube radius (= 9), and $c \approx 0.35$ is a flattening factor reflecting the torus's thinness relative to its diameter.

4.8 Radiation Belts

Jupiter's magnetosphere traps energetic charged particles (electrons and protons) in donut-shaped belts analogous to Earth's Van Allen belts but far more intense. Three nested belt zones are rendered at scaled radii:

$$r_{\text{inner}} = 1.40 R, \quad (60)$$

$$r_{\text{middle}} = 1.85 R, \quad (61)$$

$$r_{\text{outer}} = 2.35 R. \quad (62)$$

Each is an elliptically compressed torus (flattening $c \approx 0.30$ – 0.40), reflecting the belt's confinement to the magnetic equatorial plane.

4.9 Auroral Phenomena

Jovian aurorae are powered primarily by the *co-rotation breakdown* mechanism: as the magnetosphere extends beyond its co-rotation radius, field-aligned currents accelerate electrons into the polar atmosphere, producing ultraviolet and infrared emission at high latitudes. Secondary contributions come from Io's flux tube footprint.

The auroral oval is centred ≈ 15 – 20 from the pole (polar distance $p \approx 17$) and rendered as a Gaussian ribbon:

$$A_{\text{oval}}(\varphi, \lambda) = \exp \left[- \left(\frac{p - C_a(\lambda)}{w_a} \right)^2 \right] \quad (63)$$

where $C_a(\lambda) = 17 + 2.5 \sin(3\lambda + 1.2) + 1.5 \sin(5\lambda - 0.8) + 0.8 \sin(8\lambda + 2.5)$.

In the 3-D version (`DetailedModel0fJupiter.scad`), aurorae are rendered as conical frustums above each pole with a teal main glow overlaid by a thin red outer ring produced by `difference()` of two coaxial cylinders.

4.10 Tidal Interaction

Tidal interaction between Jupiter and its moons results in:

- **Tidal heating** of Io (maintaining its volcanism), Europa (maintaining its subsurface ocean), and to a lesser degree Ganymede.
- **Tidal locking** of all Galilean moons (same face always pointing toward Jupiter).
- **Gradual orbital evolution** of the system.

In the model, tidal interaction is visualised as dashed radial lines (`tidal_dash`) connecting Jupiter to each Galilean moon, co-rotating with each satellite's animated orbital angle.

4.11 Orbital Inclination of Irregular Moons

Irregular moons are thought to be captured small bodies. Prograde irregular moons (Himalia group) have inclinations 27–29; retrograde groups (Pasiphae, Ananke, Carme, Ananke) have inclinations 145–165 to the ecliptic. In the models, the orbit rings for each irregular moon are rotated by their respective inclination angles using the `rotate(inclination)` transform.

5 Model Structure and Architecture

5.1 JovianSystem.scad

The simplest file. It provides a top-level overview with:

- One global scale factor `SCALE = 0.02`.
- Pre-defined orbital radii for 8 moons and 4 rings as named constants.
- Named colour constants for all bodies (RGB triples, normalised to $[0, 1]$).
- Five primitive modules: `orbit_ring_visible`, `orbit_ring`, `moon`, `label`, and `jupiter`.
- A top-level `jovian_system()` module that assembles all parts.

Moons are placed by `rotate([0,  $\theta$ , 0])` followed by `translate([r, 0, 0])`, scattering each moon in a distinct viewing direction ($\theta = 30, 45, 60, \dots$).

5.2 ModelOfJupiter.scad

Dedicated to the surface model of Jupiter alone. Architecture:

1. **Global parameters:** $R = 40$, LAT, LON (adaptive to preview/render mode).
2. **Colour constants:** 16 named RGB vectors covering the palette from DARK to WHITE, RED_DARK to RED_LIGHT.
3. **Utility functions:** `clamp01`, `smooth01`, `mix_color`, `wrap_lon`.
4. **Noise functions:** `noise_large`, `noise_medium`, `noise_fine`.
5. **Belt/filament functions:** `wind_wave`, `ribbon`, `dark_belts`, `bright_belts`, `dark_filaments`, `bright_filaments`.

6. **Polar functions:** `polar_distance`, `polar_blend`, `polar_warp`, `polar_micro`, `polar_ring`, `polar_break`, `broken_polar_ring`, `polar_band_1-9`, `polar_filament`, `polar_filament_field`, `polar_crossflow`, `polar_vortex`, `polar_vortices`, `polar_color`.
7. **Storm functions:** `storm`, `storm.field`.
8. **GRS functions:** `spot_distance`, `spot_mask`, `spot_texture`, `apply_red_spot`.
9. **Integration:** `jupiter_color(lat,lon)` — the master colour function.
10. **Geometry:** `sphere_point`, `patch`, `jupiter` (the tessellated sphere module).

5.2.1 Piecewise Spherical Surface Approximation

The sphere is tessellated as $N_\phi \times N_\lambda$ quadrilateral patches:

$$\varphi_1(i) = -90 + \frac{180i}{N_\phi}, \quad \varphi_2(i) = -90 + \frac{180(i+1)}{N_\phi} \quad (64)$$

$$\lambda_1(j) = -180 + \frac{360j}{N_\lambda}, \quad \lambda_2(j) = -180 + \frac{360(j+1)}{N_\lambda} \quad (65)$$

Each patch is a polyhedron with four vertices (Eq. (??)) and one quadrilateral face, coloured by evaluating `jupiter_color` at the patch centroid $(\bar{\varphi}, \bar{\lambda})$:

$$\bar{\varphi} = \frac{\varphi_1 + \varphi_2}{2}, \quad \bar{\lambda} = \frac{\lambda_1 + \lambda_2}{2} \quad (66)$$

Resolution is set adaptively:

Table 3: Tessellation resolution.

Mode	N_ϕ	N_λ	Total faces
Preview	70	140	9,800
Render	170	340	57,800

5.3 DetailedModelOfJupiter.scad

This file inherits all functions from `ModelOfJupiter.scad` and adds:

1. **Feature flags** (boolean variables): `show_rotation_axis`, `show_magnetic_field`, `show_plasma_torus`, etc.
2. **Axial tilt:** `AXIAL_TILT = 3.13`.
3. **Orbital radii:** `ORBIT_IO = 80`, `ORBIT_EUROPA = 110`, `ORBIT_GANYMEDE = 150`, `ORBIT_CALLISTO = 200`.
4. **Additional polar functions** (augmenting the base polar model): `auroral_oval`, `auroral_edge`, `auroral_bands`, `central_cyclone`, `spiral_arms`, `polar_turbulence`, `ns_cyclones`.

5. **Geometry modules:** `rotation_axis_mod`, `equatorial_plane_mod`, `ecliptic_plane_mod`, `rotation_arrow_mod`, `magnetic_field_mod`, `magnetosphere_mod`, `plasma_torus_mod`, `radiation_belts_mod`, `aurorae_3d_mod`, `roche_limit_mod`, `rings_mod`, `galilean_moons_mod`, `tidal_lines_mod`.
6. **Assembly:** `jupiter_system()`.

5.3.1 Magnetic Field Line Construction

The function `dipole_pts(L,  $\phi$ , n)` generates a list of Cartesian points along a single dipole field line:

```
function dipole_pts(Lsh, phi, n=36) = [
  for (i=[0:n])
    let(
      theta = 8 + i*164/n,
      rr     = Lsh * R * pow(sin(theta), 2),
      x      = rr*sin(theta)*cos(phi),
      y      = rr*sin(theta)*sin(phi),
      z      = rr*cos(theta)
    ) [x, y, z]
];
```

The points are connected by `polyline()`, which creates a chain of `hull()` pairs between consecutive small spheres.

5.3.2 Ring System Construction

Rings are annular discs produced by differencing two coaxial cylinders:

```
module ring_torus(ir, or, h) {
  difference() {
    cylinder(h=h, r=or, center=true, $fn=64);
    cylinder(h=h+1, r=ir, center=true, $fn=64);
  }
}
```

5.3.3 Plasma Torus

```
module plasma_torus_mod() {
  color([0.75,0.25,0.55,0.14])
  rotate_extrude($fn=64)
  translate([ORBIT_I0,0,0])
  scale([1,0.35])
  circle(r=9, $fn=16);
}
```

The `scale([1,0.35])` compresses the circular cross-section vertically, producing the characteristically flat torus shape.

5.3.4 3-D Aurorae

The north auroral glow is a cone rendered as a cylinder (with non-zero base radius), while the red outer edge is a thin annular cone produced by differencing two concentric cylinders:

```
// north red edge
difference() {
  cylinder(h=R*0.14, r1=R*0.46, r2=R*0.34, $fn=FN);
  cylinder(h=R*0.14+0.1, r1=R*0.38, r2=R*0.26, $fn=FN);
}
```

5.4 MajorMoonsOfJupiter.scad

This file contains a complete Jupiter surface model plus a generic `moon()` module that encapsulates:

- An orbit ring (via the `orbit()` module in 2-D, subsequently extruded or rendered as a flat polygon chain).
- A coloured sphere at the specified position.
- A dual-line label (moon name + orbital period).

The orbit construction algorithm (Eq. (??)–(??)) is encapsulated in a module that iterates i from 0 to $N - 1$, rotating and translating a small rectangle at each step.

Moons included: Metis (0.30 d), Adrastea (0.30 d), Amalthea (0.50 d), Thebe (0.67 d), Io (1.77 d), Europa (3.55 d), Ganymede (7.15 d), Callisto (16.69 d), Himalia (250.6 d), Elara (259.6 d), Pasiphae (743.6 d), Ananke (629.8 d).

5.5 SmallerMoonsOfJupiter.scad

The most expansive catalogue. Structure:

- Full Jupiter surface model (same functions, compacted style).
- Each moon is rendered as a standalone `sphere()`, positioned by `translate([r, offset_y, offset_z])`, giving each moon a slightly distinct radial and out-of-plane position for visual clarity.
- Each moon's orbit is defined by its own named module (`orbitOfMoon()`), using the same circumference/segment formula with moon-specific inclination applied via `rotate([inclination, 0, 0])`.
- Distinct colours are assigned to each moon's sphere, distinguishing bodies that share similar orbital radii.

Moons included (partial list): Euporie, S/2003 J 18, Eupheme, S/2021 J 3, S/2010 J 2, S/2016 J 1, Mneme, Euanthe, S/2003 J 16, Harpalyke, Orthosie, Helike, and many more.

6 Geometric Construction Techniques Summary

Table 4: OpenSCAD geometric primitives and constructions used.

Feature	Primitive/operation	Notes
Jupiter surface	<code>polyhedron</code> ($\times N_\phi N_\lambda$)	Quadrilateral patches; colour per face
Moon spheres	<code>sphere(r=radius)</code>	Colour via <code>color()</code>
Orbit paths	Segmented <code>square</code> chains	$N = 180\text{--}500$ segments; circumference formula
Ring discs	<code>difference(cylinder, cylinder)</code>	Annular cross-section
Halo/main ring (simple)	<code>cylinder</code> (rotated)	Tilted slightly for viewing
Plasma torus	<code>rotate_extrude(circle)</code>	Scaled cross-section
Radiation belts	<code>rotate_extrude(circle)</code>	Elliptical scaling
Rotation arrow	<code>rotate_extrude arc</code> + cone	Partial angle extrusion
Rotation axis	<code>cylinder</code> + cones	Arrowheads at both poles
Magnetic field lines	<code>hull(sphere, sphere)</code> chain	<code>polyline</code> helper
Magnetosphere	<code>scale sphere</code>	$[2.8, 1, 1]$ — prolate in x
3-D aurorae	<code>cylinder</code> + <code>difference(cylinder)</code>	Cone + annular ring
Roche limit	<code>rotate_extrude(circle)</code> (dotted)	Six great-circle traces
Tidal lines	Radial <code>hull(sphere)</code> dashes	Co-rotating with moons
Labels	<code>linear_extrude(text)</code>	3-D extruded glyphs

7 Equation Reference Table

Table 5: Summary of all mathematical equations used in the project.

Concept	Equation
Clamp	$\text{clamp}(x, 0, 1) = \max(0, \min(1, x))$
Smoothstep	$S(x) = q^2(3 - 2q), \quad q = \text{clamp}(x, 0, 1)$
Linear interp.	$\text{mix}(\mathbf{a}, \mathbf{b}, t) = \mathbf{a} + q(\mathbf{b} - \mathbf{a})$
Angle wrap	$\text{wrap}(\lambda) : \lambda \pm 360$
Spherical coords	$(x, y, z) = (R \cos \varphi \cos \lambda, R \cos \varphi \sin \lambda, R \sin \varphi)$
Sphere eqn	$x^2 + y^2 + z^2 = R^2$
Fourier noise	$N = \sum_k A_k f_k(\alpha_k \lambda + \beta_k \varphi + \phi_k)$
Wind wave	$W(\lambda, \phi) = \sum_{k=1}^4 A_k \sin(\alpha_k \lambda + \beta_k \phi)$
Ribbon centre	$C'(\lambda) = C_0 + A \sin(f\lambda + \phi) + 0.35W(\lambda, \phi)$
Ribbon Gaussian	$R = \exp[-(\delta/w)^2], \quad \delta = \varphi - C'$
Polar distance	$p = 90 - \varphi $
Polar blend	$B_p = S(\text{clamp}((52 - p)/16, 0, 1))$
Polar ring	$R_p = \exp[-(p - C_r)^2/w_v^2]$
Ring break	$B_r = R_p \cdot Q(\lambda, \phi), \quad Q \in [0.20, 1.00]$
Polar vortex	$V = s \exp[-(d_v/\sigma)^2], \quad d_v = \sqrt{x_v^2 + y_v^2}$
Storm distance	$d_s = \sqrt{((\varphi - \varphi_c)/a)^2 + ((\lambda - \lambda_c)/b)^2}$
GRS ellipse	$d_{\text{GRS}} = \sqrt{(\Delta\lambda/18)^2 + (\Delta\varphi/7.8)^2}$
Orbit circ.	$C = 2\pi r$
Segment length	$\ell = 2\pi r/N$
Angular step	$\Delta\alpha = 360/N$
Kepler III	$T^2 = \frac{4\pi^2}{GM_J} a^3$
Laplace res.	$n_{\text{Io}} : n_{\text{Eu}} : n_{\text{Ga}} = 4 : 2 : 1$
Dipole L-shell	$r = L R_J \sin^2 \theta$
Roche limit	$d_R = 2.44 R_J (\rho_J / \rho_m)^{1/3}$
Torus surface	$(x, y, z) = ((r_0 + a \cos \phi) \cos \theta, (r_0 + a \cos \phi) \sin \theta, ca \sin \phi)$
Axial tilt	$\varepsilon = 3.13$

8 Conclusion

The *Moons & Rings of Jupiter* OpenSCAD project demonstrates that a substantial body of planetary science can be encoded in pure procedural geometry without any simulation engine or texture-mapping infrastructure. The key mathematical tools are:

- **Trigonometric Fourier-like series** for multi-scale atmospheric texture synthesis

(belts, zones, filaments, polar structures).

- **Gaussian functions** as spatial basis functions for atmospheric ribbons, polar rings, storms, and vortices.
- **Cubic smoothstep** for artefact-free transitions between atmospheric zones.
- **Linear colour interpolation** as the assembly mechanism for a multi-stage colour synthesis pipeline.
- **Spherical coordinate parameterisation** enabling all surface features to be expressed as functions of latitude and longitude.
- **Elliptical distance metrics** for correctly shaped storm ellipses and the Great Red Spot.

The physics modelled spans orbital mechanics, mean-motion resonances, planetary magnetic dipole fields, tidal interactions, plasma physics (Io torus), radiation belt geometry, and auroral emission patterns — all represented as static geometric approximations faithful to the real Jovian system at the level of visual plausibility.

The five-file hierarchy allows the project to scale gracefully from a quick visual overview (`JovianSystem.scad`) to a richly annotated and physically annotated full system view (`DetailedModelOfJupiter.scad`), demonstrating that OpenSCAD's functional programming paradigm and parametric geometry primitives are well-suited to scientific visualisation tasks of this complexity.